

NATIONAL UNIVERSITY OF SINGAPORE

FERROELECTRIC NANOSHEET FET

EE5502 : MOS DEVICES

Literature Review

A report submitted to

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1. Introduction

The motivation to start a geometric revolution starting from Planar MOSFET->FinFET->Fe-GAA FET with the aim to improve electrostatics and performance along with shrinking sizes. We observe that the combination of Ferroelectric Nanosheet FET results in perfect electrostatics and scalability due to the GAA geometry and adds new functionality: non-volatile polarization and voltage gain due to the addition of ferroelectric gate. We look into various methods to precisely control the ferroelectric properties in the Nanosheets using various stimuli in different materials. Various applications of Fe-FETs in nanoscale are also discussed such as signal followers, frequency doublers, and phase shifters, and also different devices that are modelled to overcome the limitations of traditional transistors and conventional FETs.

2. Evolution of Transistors Architectures

- As per the Moore's law: number of transistors on an integrated circuit doubles approximately every 2 years, leading to increase in computing power and a decrease in cost.
- As channel length L_G goes below 30 nm, **Dennard scaling breaks down** — the power density no longer stays constant. Shrinking dimensions causes **short-channel effects (SCEs)**: drain fields penetrate the channel, degrading threshold voltage (V_T), increasing leakage and static power consumption.
- New designs aim to **improve electrostatics rather than just shrink size**. This challenge started a geometric revolution [1]: **Planar MOSFET->FinFET->GAA->Fe-NS FET**
- Each step improves **gate-channel control** and lowers leakage.

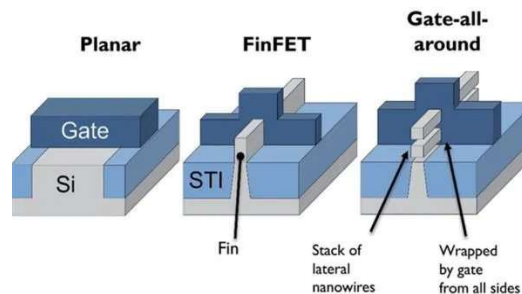


Figure 1 FET evolution

i. Planar FET:

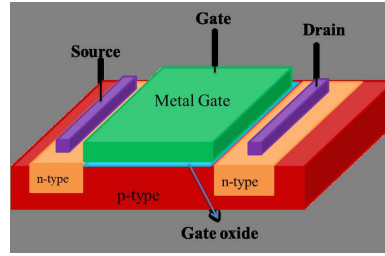


Figure 2 Planar FET

The planar MOSFET uses a flat channel and a single gate on top of SiO₂. It dominated >45 nm technology. But the **single-sided control** gets insufficient when $L_G \downarrow$. The gate potential starts decaying into the channel with a characteristic length, [2]

$$\lambda = \sqrt{\frac{\epsilon_{Si} t_{Si} t_{ox}}{\epsilon_{ox}}} \quad \text{Large } \lambda \rightarrow \text{poor control} \rightarrow \text{SCEs}$$

The **subthreshold slope**: $SS = (\ln 10) \frac{kT}{q} \left(1 + \frac{C_{dep}}{C_{ox}}\right) \approx 90 \text{ mV/dec}$ is higher than the 60 mV/dec Boltzmann limit, so leakage current increases exponentially. DIBL often exceeds 100 mV/V. So, planar devices fail below 30 nm due to higher leakage currents. [2]

ii. FinFET (First 3D Solution):

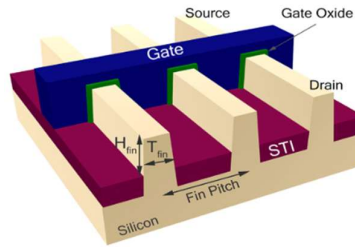


Figure 3 FinFET

To tackle SCEs, Intel and other semiconductor tech giants adopted FinFETs, a thin vertical “fin” acts as the channel and the gate wraps around three sides. This increases the effective gate area and reduces λ to

roughly[3],

$$\lambda_{Fin} \approx \frac{t_{fin}}{\pi} \sqrt{\frac{\epsilon_{Si}}{\epsilon_{ox}}}$$

Better control yields $SS \approx 70 \text{ mV/dec}$, $DIBL \approx 60 \text{ mV/V}$, and $I_{on}/I_{off} \sim 10^6$. FinFETs enabled 22–10 nm nodes and became the mainstream 3-D transistor.

FinFET Limits and GAA Transition:

- Below 5 nm, fin-aspect limits and leakage at fin edges degrades performance due to **corner-field crowding** as the gate doesn't fully surround the channel. Tall fins complicate lithography and heat removal. [4]
- Gate-All-Around (GAA) devices fix this by wrapping the gate on all sides, giving symmetric control and uniform current. At equal L_G , GAA reduces DIBL by $\sim 30\%$ and achieves $SS \approx 60$ mV/dec – the theoretical limit for thermionic carriers. [4]

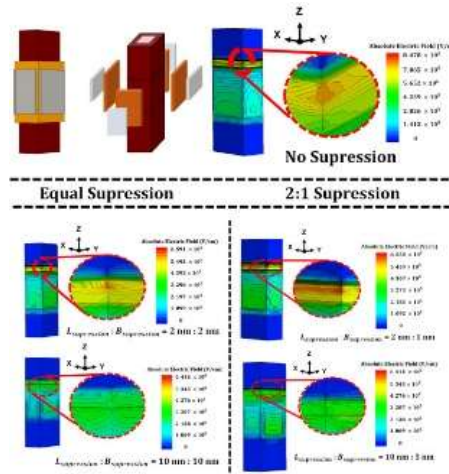


Figure 4 Corner field crowding

iii. Nanosheet FET:

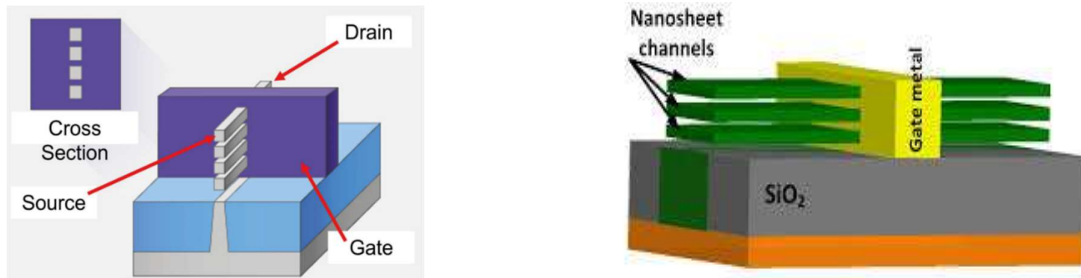


Figure 5 Nanosheet FET

The Gate-All-Around (GAA) design surrounds the channel completely, giving 360° electrostatic control and enabling further scaling. [5]

Key Features: 3–5 stacked Si/SiGe sheets $\rightarrow$ higher drive current (I_D) per footprint. High- κ $\text{HfO}_2/\text{SiO}_2$ dielectric and metal gate wrap around each sheet. [5]

Current Scaling [5]: $I_D \propto N_{\text{sheet}} \times (W_{\text{sheet}} \cdot \mu \cdot C_{\text{ox}})$

More sheets = higher drive current without larger area

Electrostatic Integrity [5]: $\lambda = \sqrt{\frac{\epsilon_{si}}{\epsilon_{ox}}} t_{si} t_{ox}$ Smaller λ ($\approx 3-5$ nm) $\rightarrow$ better short-channel control

Challenges: Complex epitaxy and selective etch and GAA being a Logic-only device there is no data retention.

iv. Ferroelectric FET (FeFET):

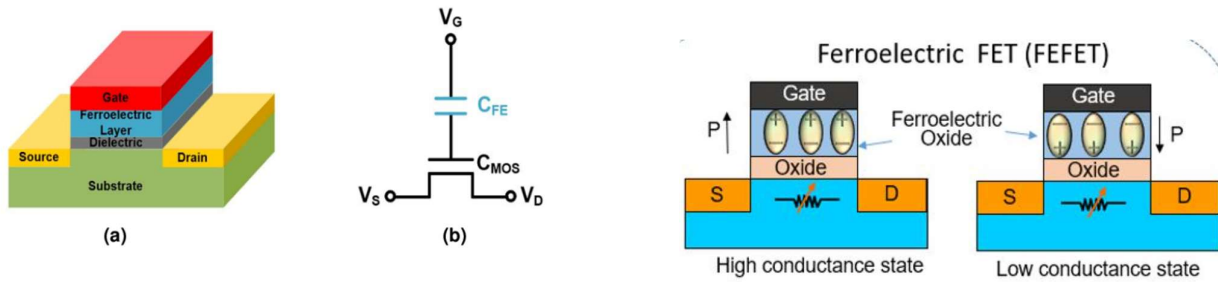


Figure 6 Nanosheet FET

Concept: (i) A FeFET replaces the gate with hafnium-oxide based ferroelectric, storing charge via remanent polarization ($\pm P_r$). [6]

(ii) When voltage is applied, dipoles align, creating two stable polarizations $\pm P_r$. These shift the threshold voltage: logic “1” = $+P_r$, “0” = $-P_r$ [6]

Physics: (i) Ferroelectric behavior follows the Landau–Devonshire model:

$$F(P) = \alpha P^2 + \beta P^4 + \gamma P^6 - EP$$

(ii) Negative curvature in $F(P) \rightarrow$ negative capacitance region $\rightarrow$ voltage amplification $\rightarrow$ steep SS slope ≈ 60 mV/dec.

Key Points: Advantages: Non-volatile logic, faster switching, lower VDD ($\sim 0.3-0.5$ V).

Limitations: Retention loss, interface traps, weaker electrostatics in planar/FinFET.

v. Ferroelectric Nanosheet FET (Fe-NS FET):

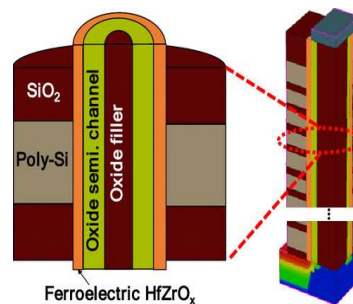


Figure 7 Fe-NS FET

Features:

- Metal Gate & Nanosheet Channel: provides work-function tuning, high mobility Si/SiGe stacked layers fully enclosed by gate.
- Ferroelectric Layer: switchable polarization ($\pm P_r$)
- Interfacial Oxide: ensures stability & leakage suppression

Combining both GAA and Ferroelectric provides:

- GAA geometry $\rightarrow$ perfect electrostatics & scalability
- Ferroelectric gate $\rightarrow$ new functionality: non-volatile polarization & voltage gain

3. Ferroelectric Materials

FeFETs use ferroelectric materials as gate oxide insulator to achieve polarization states of the ferroelectric layer. However, choice of the thin material is important for performance, scalability and compatibility with present topologies such as CMOS. The main kinds of ferroelectric materials used are Hafnium oxide based, Pervoskite Oxides, organic Ferroelectrics and two dimensional ferroelectrics (evolving).

In the past decade, research on various fronts is being carried out in ferro-electronics especially on scalability of memory devices to nanoscale. To properly bring out their advantages for the given application, it is important to know the material's properties under various conditions. The recent advancements in SMO compounds and their multiferroic control are discussed here.

- SMO based compounds such as SrMnO_3 , a perovskite multifunctional oxide material are ideal candidates for next generation memory. [7]
- They have strong ferroelectric behaviour and exhibit spontaneous polarization which can be reversed when electric field is applied. This also allows to store large amount of data per volume.
- They also exhibit ferromagnetic properties which helps us to encode data in both electric and magnetic states enhancing functionality and speed.
- The main challenge is maintaining or optimizing the multiferroic properties as material dimensions reach nanoscale.

- It is crucial to understand the material properties under external stimuli (temperature, pressure, strain, electric fields), over dimensionality (nanosheets) and internal composition (doping, defects, strain).

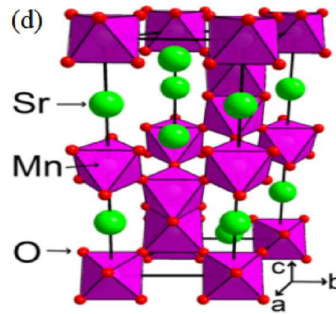


Figure 8 6H SMO compound

i. Control strategies of multiferroicity

Ferroelectric materials respond to the following stimuli in different ways like temperature, pressure, strain, process variations and dopants which can be used in various applications:

Temperature, Pressure, Strain variations [7]

- Phase transitions of these compounds are influenced by change in temperature and pressure.
- Below 1035 °C in air, SMO is a stable four-layer hexagonal phase. As the temperature increases, oxygen content gradually reduces and forms vacancies which acts as charge carriers.
- Oxygen deficiency also results in the enhancement of ferromagnetic coupling ($\text{Mn}^{3+} - \text{Mn}^{3+}$).
- The ferroelectricity of SMO is mainly induced by epitaxial strain. SMO films grown on SrTiO_3 (STO) substrates are strained due to lattice mismatch, inducing ferroelectricity.
- Its ferroelectricity and magnetism can be further regulated by doping (such as Ca and Ba) or interface engineering.

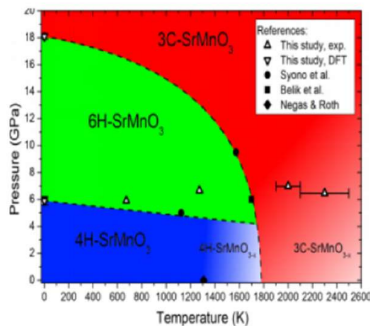
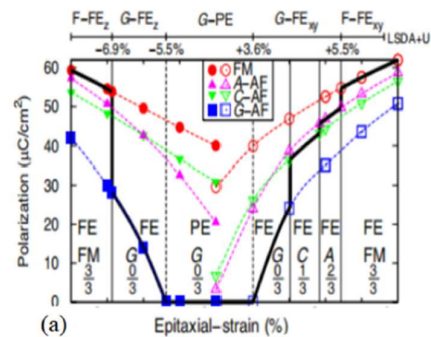


Figure 9 Phase transitions Figure



(a) 10 Effect of strain on Polarization

Epitaxial Growth Techniques [7]

Vast amount of research is being carrying out for tailing properties through growth control on thin films.

- Pulsed Laser Deposition: PLD is one of the most popular methods for fabricating SMO thin films. Here, the SMO target is exposed to high-energy laser pulses, which evaporate the surface material and generate plasma. It can be used to fabricate high-quality epitaxial films on a range of substrates because it can accurately control the thickness and composition of the film.

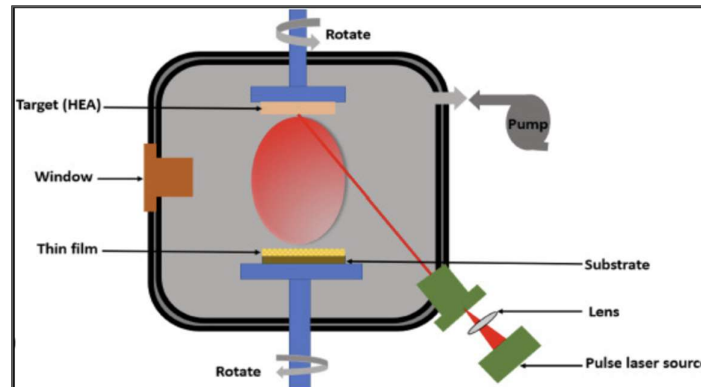


Figure 11 PLD setup

- Molecular Beam Epitaxy: MBE is used to obtain precise atomic layer growth with very good crystal quality. The substrates fabricated using MBE have an average thickness of 21.6 nm with a good control over tensile strain, polarization and magnetic properties.

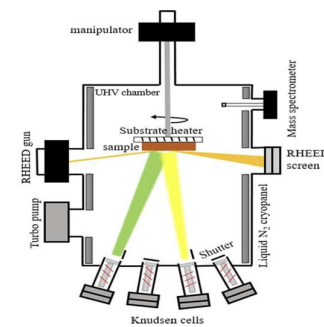


Figure 12 MBE Setup

- Magnetron Sputtering: This method is used to prepare thin films with high dielectric constants (K). The principle of sputtering involves ion bombardment on a target. They offer low temperature processing and controlled growth.

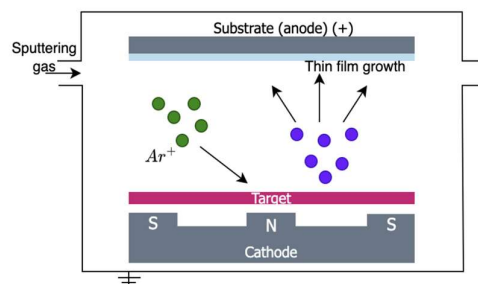


Figure 13 Magnetron Sputtering Deposition

Other materials such as CuSrS₂ based nanosheets are being manufactured due to the limitation of PbTiO₃ and BaTiO₃ having low ferroelectricity in nanoscale devices. Their ferroelectricity becomes weak or even disappears when their thickness reaches the nanoscale due to the depolarization field. This takes us to research on synthesis of 2D CuCrS₂ nanosheets with thicknesses down to one unit cell via van der Waals epitaxy in a chemical vapor deposition (CVD) system. The crystal structure of CuCrS₂ is non-centro symmetric, which can be regarded as inserting Cu atoms between 1T-CrS₂ layers. The key to this process is utilizing freshly cleaved mica sheets, which have an atomically smooth and chemically inert surface, as the van der Waals epitaxial substrates. [8] A 1.8 nm nanosheet was synthesized using van der Waals epitaxy via CVD which can be used for scalable synthesis.

Ferroelectricity was directly measured by piezo response force microscopy (PFM) at room temperature in samples as thin as 10 nm. The material has a high curie temperature of approximately 700 K. The fabricated nanosheet exhibited a robust and stable performance proving its potential for future 2D ferroelectric memory and computing.

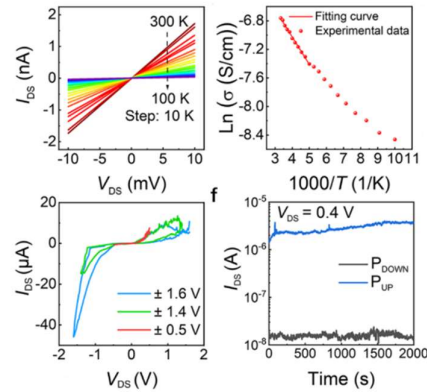


Figure 14 I-V, conductivity-Temperature and endurance characteristics of FSJ device [8]

While considering organic ferroelectric materials such as lead zirconate titanate (PZT) ceramics show positive strain under electric field, piezoelectric polymer polyvinylidene fluoride (PVDF), an inorganic ceramic filler exhibits negative electro-strain. [9]

The reason behind this is still under research, however, this unique property of this material is also an avenue explore. The addition of dopants like PVDF has improved the control of negative strain by about 91% which in turn increased the polarization by about 22%. The reason behind this negative

strain property of PVTF along with PTO nanosheets is the interaction between them altering their molecular chain arrangement in the amorphous phase enhancing contraction effect. There is also an increase in the degree of crystallinity (X_c) is observed in the doped nanosheet indicating that in-situ electrostatic field is very effective thereby increasing the controllability of the ferroelectric device.

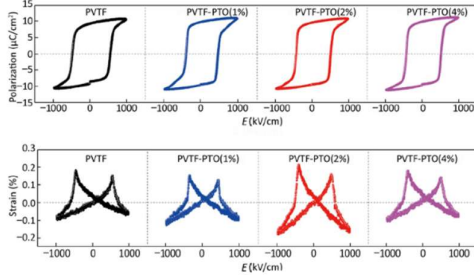


Figure 15 Effect of electric field on polarization and strain

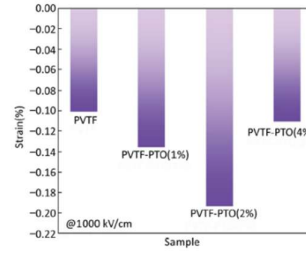


Figure 16 Negative strain values

4. Limitations of Present Devices:

Traditional transistors can typically perform only a **single function at a time**, which restricts the versatility in multifunctional circuits. Schottky FETs, even though they are useful, suffer from **high leakage currents**, and realizing them for **multifunctional operations** demands multiple transistors, which increases area cost as well as circuit complexity. As the technology is growing, CMOS scaling has slowed down, limiting further improvements in **transistor performance**, energy efficiency, and integration density.

On the other hand, Conventional FETs struggle to achieve **subthreshold slopes below 60 mV/dec**, making it difficult to design low-voltage, energy-efficient systems, and for emerging fields like AI and neuromorphic computing require **compact, low-power, multi-level devices**—driving interest in ferroelectric-CMOS integration, which offers potential but demands careful design and optimization.

5. Technological Advancements:

The proposal of an n-TYPE Germanium-based Fe-TFET with a **hafnium-oxide-based ferroelectric ($\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$) layer** [10] has shown a breakthrough performance for low-power analog operations, as it can perform many operations, such as a **frequency doubler, phase shifter, signal follower**, along with the modulation of V_{th} . This Fe-TFET works under the principle of band-to-band tunneling currents, where (I_{sc}

is Source-to-channel current and I_{CD} is channel-to-drain current). Ferroelectric polarization modulates the tunneling barrier height, which helps in shifting the threshold voltage (V_{th}). By programming the ferroelectric polarization with short voltage pulses, **namely V_{PROG}** , the device can switch between frequency doubler, phase shifter, and signal follower modes.

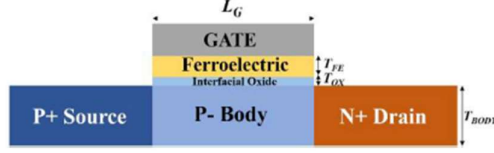


Figure 17 Schematic of n-type Ge based TFET with ferroelectric oxide layer

When $V_{PROG} = 0$ V, $I_{SC} = I_{CD}$ will be present (i.e., Symmetric currents), which produce an **output twice the input frequency**. So effectively it works as a frequency doubler which operates across 1-10MHz with minimal higher-order harmonics. When $V_{PROG} = +4$ V, we would observe an upward polarization, which makes I_{SC} dominate. The output waveform will be inverted, resulting in a **180-degree phase shift**. So, it acts as a Phase shifter. When $V_{PROG} = -4$ V, we would observe downward polarization, which makes I_{CD} dominate. The output waveform follows the input waveform with a **0-degree phase shift**. So, it works like a signal follower. Hence, it is observed that Fe-TFET achieves low power, non-volatile, and high-frequency signal processing in a single device by dynamically controlling the tunneling barriers with ferroelectric polarization.

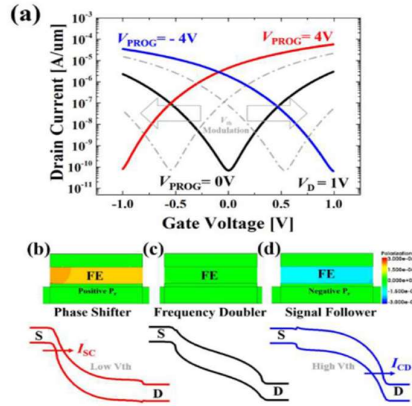


Figure 18 (a) Transfer characteristics of the Fe-TFET for $V_{PROG} = 4.0$ V, 0.0 V, and -4.0 V. (b)–(d) Polarization distributions and energy band diagrams after each programming

In Fig. (a) we can see that the **red curve** corresponds to phase shifter operation. This is shifted towards the left due to **upward polarization**. The **blue curve** corresponds to the signal follower, and the curve is shifted toward the right due to **downward polarization**. As a frequency doubler has **no change in the polarization**,

it will remain as an ideal IV curve. In Fig. (b), the band diagram of the phase shifter shows that there is a band bending at the source channel interface due to upward polarization, which lowers the barriers for charge movement. So, the charge carriers tunnel and will make I_{SC} dominate. **It gives a lower V_{th} .** In Fig(c), the band diagram of the frequency doubler seems to be symmetric throughout the device as there is no polarization. In Fig. (d), the band diagram of the Signal follower shows band bending near the drain channel interface due to downward polarization. Similar to a phase shifter, the charge tunnels will make I_{CD} dominate. **It enhances the V_{th} .**

Exploring Negative Capacitance Devices, Ferroelectric FETs, and Ferroelectric Tunneling Junctions (FTJs) [11] paved the way for **next-generation low-power and multifunctional electronic devices** by enabling **voltage amplification, polarization-controlled conduction, and ultra-low-energy memory operations.** These devices collectively address the challenges of **energy efficiency, scaling limitations, and multifunctionality** in modern transistor technologies.

Negative Capacitance (NC) Devices: The ferroelectric layer shows smooth switching **without hysteresis.** This **voltage boost (gain >1)** helps achieve a steeper subthreshold slope, allowing transistors to work efficiently at lower supply voltages.

Ferroelectric FETs (FeFETs): The polarization in the ferroelectric layer controls the channel current (I_{DS}), which enables **programmable conductance.** High channel doping keeps polarization stable but weakens current modulation, while low doping improves read current difference but needs extra charge from recombination or Schottky contacts.

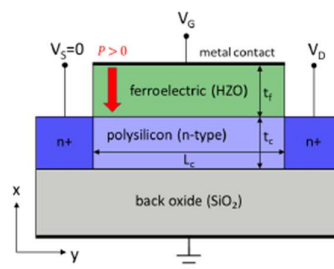


Figure 19 Crossectional view of Fe-FET

Ferroelectric Tunneling Junctions (FTJs): The tunneling current depends on **ferroelectric polarization**, which enables **ultra-low-energy memristive behavior**. MFIM structures have separate switching and tunneling, but designers need to balance oxide voltage, depolarization field, retention, and read current for optimal performance.

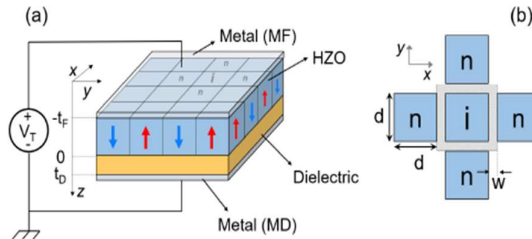


Figure 20 a) Sketch of a MFIM capacitor, b) Ferroelectric domains

6. Gaps and Opportunities for Improvement:

Current results are **simulation-based**; no real devices or prototypes have been tested, including FeFETs with low-doped channels. **Effects of thermal noise, parasitic elements, and large-scale array behavior** are not fully explored. FTJs face challenges in **retention and depolarization stability**, and the long-term reliability of NC transistors remains untested. Build and test actual device prototypes, optimizing fabrication and integration with standard CMOS technology. There is a need to test under real conditions, including thermal noise, parasitics, and large-scale array behavior, ensuring reliable operation over time.

7. Conclusion:

We have explained the evolution of Ferroelectric Nanosheets along with the device physics. We have seen how parameters such as subthreshold slope, DIBL, etc., are the reasons for the planar devices' failure below 30nm and the rise of FinFET. Ferroelectric nanosheet FET (Fe-NS FET) provides better functionality when compared to other devices. We surveyed the recent research on various ferroelectric devices, especially SMO-based components. Various ways in which external stimuli can control the ferroelectric and magnetic properties. We have also surveyed a CuCrS₂-based nanosheet, which doesn't lose its ferroelectricity in sub-nanometer lengths have seen different applications of Fe-FETs as signal followers, frequency doublers, phase shifters, and different devices such as NC Devices, Ferroelectric FETs, Ferroelectric Tunneling junctions to overcome the limitations offered by traditional transistors and FETs.

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9. Contribution

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